Supplement 3: Runtime & Abstraction Tax

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> *Every abstraction layer has a CPU cost. The question is whether that cost buys enough isolation to be worth it.*

The Boundary Leak Penalty

Zero-cost exceptions are a runtime mirage. When a `DatabaseConnectionException` is thrown across module boundaries and caught by a higher-level caller, modular encapsulation breaks. Lemire's benchmarks put the cost at $10,000\times$ the normal execution path.

This is not a language issue. It is a structural one. The CPU must flush pipelines, unwind stacks, and reconstruct state precisely because the developer forced two decoupled modules to become intimately aware of each other's internal failure modes. The performance penalty is the machine's protest against a broken boundary.

Leak types and their costs:

Boundary Leak	Manifestation	Typical Penalty
Exception cross-boundary	Stack unwind, pipeline flush	$10,000\times$ baseline
Direct buffer bypass	Cache miss chain	$10\times$ per access
Implicit heap allocation	GC pressure, allocation stalls	Variable, cumulative

The Useful Fiction of the Stream Abstraction

```
BufferedInputStream bis = new BufferedInputStream(new FileInputStream("file.dat")
```

The abstraction layer hides: OS buffer allocation, disk sector alignment, interrupt coalescing, page cache lookups, DMA transfer scheduling. This is deep modularity working as designed.

When it breaks:

```
FileChannel channel = FileChannel.open(Path.of("file.dat"));
ByteBuffer buffer = ByteBuffer.allocateDirect(1024 * 4);
```

Zero-copy bypass. Manual buffer management. No abstraction between your code and the hardware. The stream fiction collapses because the system cannot afford the translation layer.

The architectural question is not whether to use buffers or streams. It is whether the boundary you drew can survive contact with production load. If a clean abstraction imposes a $100\times$ penalty at 10,000 requests per second, the abstraction was misdesigned for its runtime context.